Combined probabilities

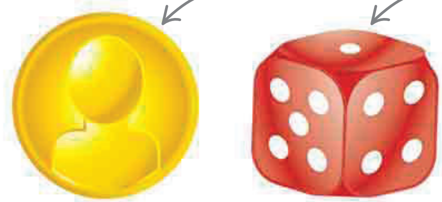
THE PROBABILITY OF ONE OUTCOME FROM TWO OR MORE EVENTS HAPPENING AT THE SAME TIME, OR ONE AFTER THE OTHER.

SEE ALSO
❮ 230–231 What is probability?
❮ 232–233 Expectation and reality

Calculating the chance of one outcome from two things happening at the same time is not as complex as it might appear.

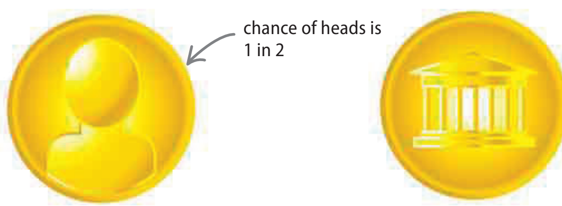
What are combined probabilities?

To find out the probability of one possible outcome happening from more than one event, all of the possible outcomes need to be worked out first. For example, if a coin is tossed and a dice is rolled at the same time, what is the probability of the coin landing on tails and the dice rolling a 4?



Coin and dice
A coin has 2 sides (heads and tails) while a dice has 6 sides—numbers 1 through to 6, represented by numbers of dots on each side.

▷ **Tossing a coin**
There are 2 sides to a coin, and each is equally likely to show if the coin is tossed. This means that the chance of the coin landing on tails is exactly 1 in 2, shown as the fraction ½.

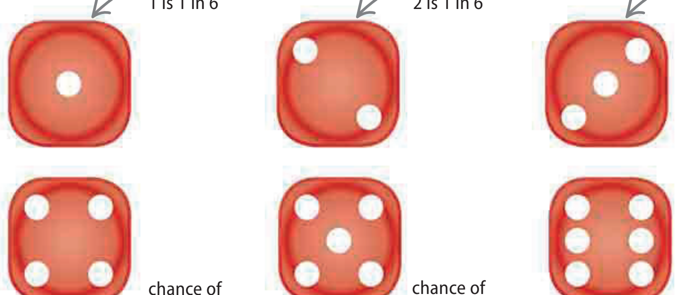


$$\frac{1}{2}$$

represents chance of single event, for example chance of coin landing on tails

represents total possible outcomes if coin is tossed

▷ **Rolling a dice**
Because there are 6 sides to a dice, and each side is equally likely to show when the dice is rolled, the chance of rolling a 4 is exactly 1 in 6, shown as the fraction ⅙.



$$\frac{1}{6}$$

represents chance of single event, for example chance of rolling a 4

represents total possible outcomes if dice is thrown

▷ **Both events**
To find out the chances of both a coin landing on tails and a dice simultaneously rolling a 4, multiply the individual probabilities together. The answer shows that there is a 1⁄12 chance of this outcome.

$$\frac{1}{2} \times$$

$$\frac{1}{6} = \frac{1}{12}$$

multiply the 2 probabilities together

chance of coin landing on tails is 1 in 2

chance of dice rolling a 4 is 1 in 6

chance of specific outcome

chance of coin landing on tails and rolling a 4 is 1 in 12

total possible outcomes